

Project Assignment 1

Classifying AD patients and normal controls from brain images

Purpose:

This project is for AD prediction. Students are expected to be applying what they have learnt in Lab 1 about brain image processing and in Lab 2 about medical image classification to complete this project. This is a group-work based project.

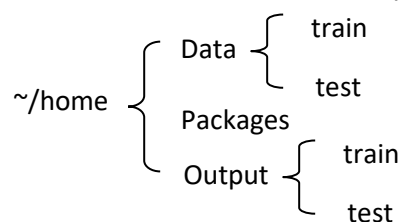
Data:

There are 40 subjects (being either AD patients or normal controls) in the training set and 10 subjects in the test set. The T1-weighted MR brain images for all subjects are provided. In addition, the template images, and the brain AAL atlas could be found under the “Packages” directory.

Tasks:

Predict whether each of the 10 subjects in the test is an AD patient or not. This includes extracting the feature representation from each subject’s MR brain image and classifying these feature vectors using Support Vector Machines.

For feature extraction, students have practiced how to measure the grey matter volume in a single ROI in Lab 1. Now you need to extend the measurement to the first 90 ROIs in AAL atlas for each subject. Therefore, you may do this by using unix shell programming to process these data in batch. A very simple tutorial about unix shell programming is provided in Canvas. Please follow the links in that document for further reading if you need. The feature extraction process includes four sub-tasks: skull-stripping, tissue segmentation (for grey matter), registration, and measurement. To make the tasks easier, we have provided templates of shell scripts to conduct these four sub-tasks, with main steps removed. Please complete those shell scripts for your tasks. It is worth noting that, in those templates, the files are assumed to be organized in a specific directory structure. Be careful to change the directories in the code if you want to use a different directory structure.



After feature extraction, each subject is represented by a 90-D feature vector. Please use Support Vector Machine for the classification. You may find the components in the feature vectors have large values and they change largely from feature to feature. You can conduct feature normalization before you apply SVM. For example, you can normalize each feature (across all training samples) to have a zero mean and a unit standard deviation. This may help your classification.

Install the virtual machine as introduced in Lab 1.

Folder Structure

1. **elec5622** – contains the four bash files to be modified (SkullStripping, TissueSegmentation, Registration and Measurement) from Canvas->Modules->Week 7->code.zip

NOTE: *DO NOT* modify the “CreateSeedMask” file.

2. **Data** - Store raw data files (50 files including both AD patients and Normal Controls) in two subfolders (train and test)

Action: Please download Original.zip file from Canvas->Modules->Week 7. In the Data folder, you need to copy the 40 data files (Data_00.nii to Data_39.nii) unzipped from Original.zip into the train folder, the rest 10 images (Data_40.nii to Data_49.nii) into the test folder.

3. **Packages** – Store the template and the AAL atlas files downloaded from Canvas->Modules->Week 7-Packages.zip.
4. **Output** – Empty folders for you to save results.

Files Uploaded on Canvas:

1. The files in code.zip are the ones you need to focus on except the “CreateSeedMask” file. Please complete the lines after TODO in these files.
2. The Original.zip has the information you need to complete SVM task.
3. The files in Packages.zip include the template files and the AAL atlas.

Submission deadline:

This project is due on: 11:59pm, Oct 11, 2026 (Week 9 Sunday)

This project occupies 15% of the total marks.

Submissions:

Please submit (1) a complete report (more instructions could be as follows), and

(2) a spreadsheet containing the 90 ROI grey matter volumes for all 50 subjects in both the training and the test sets.

As for the report, **the team members are requested to submit a group report, and clearly state the contribution of each team member at the beginning of the report.** It is recommended to organize the report in the following way.

Section 1: Aims and Background

Section 2: Contributions of Team Members

Section 3: Methodology

Section 4: Results

Please include your prediction labels for Data_40.nii to Data_49.nii into a Table:

File Name:	Prediction
Data_30.nii	[AD/NC]
...	[AD/NC]
Data_49.nii	[AD/NC]

Section 5: Discussion and Conclusion

Evaluation criterion:

Criteria	Ratings			
Introduction: Aims and Background (20%)	20.0 pts Exceptional Defines and elucidates the problem. Develops compelling rationale for the lab. Persuasively explains the multifactorial contribution of learning gained from the lab.	15.0 pts Good Defines problem with some depth. States rationale for the lab. Provides explanation beyond basic contribution.	10.0 pts Average Defines problem. States only obvious rationale for the lab. Explains the basic contribution of the activities to the understanding of the lab.	3.0 pts Poor Shows fundamental lack of understanding. Does not state rationale for the activity. Does not explain the contribution of the activities to the understanding of the lab.
Background and Methods: Knowledge of the area and use of sources (30%)	30.0 pts Exceptional Demonstrates exceptional depth of knowledge of the field. Comprehensive use of most recent and seminal sources. Clearly discriminates among seminal sources.	25.0 pts Good Demonstrates proficient knowledge of the field. Thorough selection of sources pertinent to project. Shows some discrimination among seminal sources.	15.0 pts Average Demonstrates a basic knowledge of the field. Selected sources relevant to project. Limited discrimination among seminal sources.	7.0 pts Poor Lacks a basic knowledge of the field. Selected sources irrelevant to project. Does not discriminate among seminal sources. Misinterprets sources.
Results (20%)	20.0 pts Exceptional Results organised into sections and subsections in a logical order.	18.0 pts Good Results are well organised in a logical order.	15.0 pts Average Results organised into sections and subsections.	7.0 pts Poor Results are poorly organised.
Discussion and Conclusion (30%)	30.0 pts Exceptional Outstanding and innovative synthesis between results and sources.	25.0 pts Good Demonstrates synthesis between results and sources.	20.0 pts Average Relevant information about the results is discussed under each heading. Demonstrates information synthesis to a limited extent.	10.0 pts Poor Information on the results are repeatedly reported without any synthesis from sources.
Total Points: 100.0				